

Deployment Guide

Recommended setup

This repository now includes a production-ready Flask backend in `webbackend.py`

(`/d:/Project/DyslexiaDetectionSystem/webbackend.py`) and a Docker deployment definition in `Dockerfile`

(`/d:/Project/DyslexiaDetectionSystem/Dockerfile`).

The deployment serves:

- the standalone browser dashboard from `web/` (`/d:/Project/DyslexiaDetectionSystem/web`)
- the local transcription API at `/api/reading-transcribe`
- a health check at `/healthz`

Deployment Options

The project now includes configuration for three common hosting paths:

| Platform | File | Best for |

| Render | [render.yaml](/d:/Project/Dyslexia_Detection_System/render.yaml) | Simple

Docker deployment with automatic HTTPS |

| Railway | [railway.json](/d:/Project/Dyslexia_Detection_System/railway.json) | Fast

app deployment from GitHub |

| Fly.io | [fly.toml](/d:/Project/Dyslexia_Detection_System/fly.toml) | Global app

hosting with region control |

Live Deployment

Current public demo:

- <https://learning-disorder-intelligence.onrender.com/>

(<https://learning-disorder-intelligence.onrender.com/>)

Deploy link:

- <https://learning-disorder-intelligence.onrender.com/>

(<https://learning-disorder-intelligence.onrender.com/>)

Render

Render is the simplest path if you want a quick public URL.

Steps:

1. Push this repository to GitHub.

1. Connect the repo to Render.

1. Let Render build the Docker image.

1. Open the public URL and verify `/healthz` returns ok.

1. Test the dashboard microphone workflow from the HTTPS URL.

Render auto-deploy troubleshooting

If the site does not update after a GitHub push, check these items in the Render dashboard:

1. The service is connected to archisman-das/Learning-Disorder-Intelligence.

1. The deploy branch is main.

1. Auto-deploy is enabled for the service.

1. The latest build completed successfully.

1. The health check path is still /healthz.

This repository is already configured for automatic deploys through render.yaml

(/d:/Project/DyslexiaDetectionSystem/render.yaml), which sets autoDeploy: true and points Render at

the Dockerized Flask backend.

Railway

Railway works well if you want a quick Git-based deploy with a Dockerfile.

Steps:

1. Push this repository to GitHub.

1. Create a Railway project from the repo.

1. Railway should use railway.json (/d:/Project/DyslexiaDetectionSystem/railway.json) and Dockerfile

(/d:/Project/DyslexiaDetectionSystem/Dockerfile).

1. Confirm the service exposes the app on the assigned \$PORT.

1. Visit /healthz and then open the public dashboard URL.

Fly.io

Fly is a strong choice if you want region-aware global deployment.

Steps:

1. Install flyctl.

1. Run fly launch or reuse the included fly.toml (/d:/Project/DyslexiaDetectionSystem/fly.toml).

1. Deploy with fly deploy.

1. Confirm the app is reachable over HTTPS and /healthz responds successfully.

1. Open the dashboard and test the microphone workflow.

Important notes

- Microphone access requires HTTPS or localhost.
- The transcription endpoint depends on openai-whisper and ffmpeg.
- The first audio request may be slower because the Whisper model is loaded on demand.
- The dashboard itself is mostly client-side, so browser state and records are stored locally in the user session unless you extend the backend further.
- The local web dashboard also includes a model statistics page that reads from stored comparison snapshots rather than live test labels.
- For Fly.io, the app is configured to listen on port 10000 and expose /healthz as a health check.

Local run

For local development:

```
python run_local_web.py
```

For production-like local testing:

```
gunicorn -b 0.0.0.0:10000 web_backend:app
```